

Anindita Bhattacharjee

Graduate Student

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github.com/Anindita-Bhattacharjee-12

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RESEARCH INTERESTS

Computer Vision, Machine Learning Algorithms, AI In Healthcare, LLM Interpretability, Human-Computer Interaction

EDUCATIONAL BACKGROUND

Bachelor of Science in Computer Science and Engineering

2022–2026

Major in Information System

CGPA: 3.97

Honors: Summa Cum Laude (Gold Medalist)

American International University–Bangladesh (AIUB)

RESEARCH EXPERIENCE

Research Assistant

Jun 2026 – Present

ELITE Research Lab LLC

Research Intern

Jan 2026 – Apr 2026

Department of Deep Learning and Computer Vision

Advanced Machine Intelligence Research Lab (AMIR Lab)

RESEARCH PAPERS

- **A. Bhattacharjee**, S. A. Hamim, M. F. Mridha, and Md. J. Hossen et al. *“SpatioTrace: An Interpretable Framework for Copy-Recall Divergence Analysis in ROCF Drawings”*, Manuscript submitted to IEEE Transactions on Neural Systems and Rehabilitation Engineering (TNSRE), **under review**, 2026.
- **A. Bhattacharjee**, A. A. Hasib, Md. M. I. Hasan, S. I. Hasan, Dr. D. Nandi et al. *“From Reactive to Proactive: Multimodal Therapist Intervention Prediction for Children with Autism Spectrum Disorder”*, Undergraduate thesis manuscript, ready for submission.
- **A. Bhattacharjee**, Md. K. Morol et al. **Prospective title:** *“Hidden in the Layers: Internal Representations for Human-AI Code Authorship Attribution”*, Manuscript in preparation.

PROJECTS

- **Truthful Nudging: Auditing AI-Generated Price Justifications**
 - Tests whether AI explanations that nudge house-price estimates are truthful by comparing claimed influential features with their actual SHAP importance and compares direct price statements and critique-style questions for verifiability and explanation error.
- **Synthesizing Solutions: StarGAN v2 and WTAD-YOLO for Tomato Disease Detection**
 - Explores combining StarGAN v2 with WTAD-YOLO (tomato leaf disease detector) to generate synthetic diseased-leaf images and fix class imbalance in the disease dataset.
 - Compares the two model’s scale and diversity -StarGAN v2’s large multi-domain datasets vs WTAD-YOLO’s small, imbalanced 5-class tomato disease set to argue generative augmentation could boost detection robustness.
- **Mapping the Causal Architecture of Academic-pressure Symptoms: A Network Analysis of Anxiety, Stress and Depression across Bangladesh Private Students**

- Builds a symptom network (26 GAD-7/PSS-10/PHQ-9 items) from survey data on 1,977 students, using correlation thresholds and centrality measures to find which symptoms bridge anxiety, stress and depression.
- Applies two independent causal discovery methods - Bayesian network learning and the PC algorithm to identify directed causal pathways between symptom domains which checked with bootstrap stability analysis.
- **Breaking Barriers Through Touch: An Affordable Tactile Controller for Vision-Impaired Computer Access**
 - An Arduino based physical control panel with tactile buttons, a scroll knob, audio feedback and vibration and letting blind, low-vision or motor-impaired users operate a computer without a keyboard or mouse and two-step confirmation to prevent accidental input.
- **Spatial Decay Clustering in ROCF Copy Drawings: Peripheral vs. Central Error Signatures**
 - Measures how much each patient's copied figure deviates from the standard ROCF figure, region by region and tests whether errors cluster more at the periphery than the center and uses K-means on these spatial error patterns to split patients into high-error/low-error clusters, then applies SHAP to identify which grid regions (peripheral vs. central) drive that split.

TECHNICAL SKILLS

- **Programming Languages:** C++, Java, C#, Python, R
- **Software & Simulations:** Pytorch, TensorFlow, Scikit-learn, MATLAB
- **Data Visualization:** Matplotlib, Seaborn
- **Web Development:** HTML, CSS, JavaScript, PHP
- **Database Management:** MSSQL, MySQL, Oracle
- **AI & ML:** Machine Learning, Deep Learning, Computer Vision
- **Tools:** Git, Latex, AutoCAD

AWARDS

Dean's Honor List for Academic Excellence	<i>2023-2026</i>
Faculty of Science and Technology, American International University-Bangladesh	
<i>Awarded in: Fall 2023-24, Spring 2023-24, Fall 2024-25, Spring 2024-25, Fall 2025-26</i>	
Merit-Based Academic Scholarship	<i>2023-2026</i>
Faculty of Science and Technology, American International University-Bangladesh	
Bangladesh Govt. Education Board Junior Scholarship	<i>2015</i>
Sylhet Zone, Bangladesh	

CERTIFICATIONS

Programming Girl's Competition	<i>2024</i>
American International University-Bangladesh	

REFERENCES

Dr. Dip Nandi

Professor, Dean, Faculty of Science and Technology
 American International University-Bangladesh
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Dr. Mohammad Firoz Mridha

Professor, Head of the Department, Faculty of Science and Technology
 American International University-Bangladesh
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